

Phishing Email Analysis and IOC Extraction Report

Project: Phishing Email Analysis and IOC Extraction Workflow

Case Study: DHL Express Shipment Notification

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Classification: Phishing Email

1. Executive Summary:

This report analyzes a phishing email that impersonates DHL Express to deceive recipients into opening a malicious attachment. "The attacker leverages brand impersonation, sender impersonation, and a malicious Excel attachment to increase credibility and deliver a malware payload."

2. Scenario Overview

The email claims to be a shipment notification from DHL Express. The message attempts to create trust by using DHL branding and shipment related language. The primary objective of the attacker is to convince the recipient to open an attached Excel document that may contain malicious macros or hidden payloads.

3. Phishing Techniques Identified

Spoofed Sender Identity:

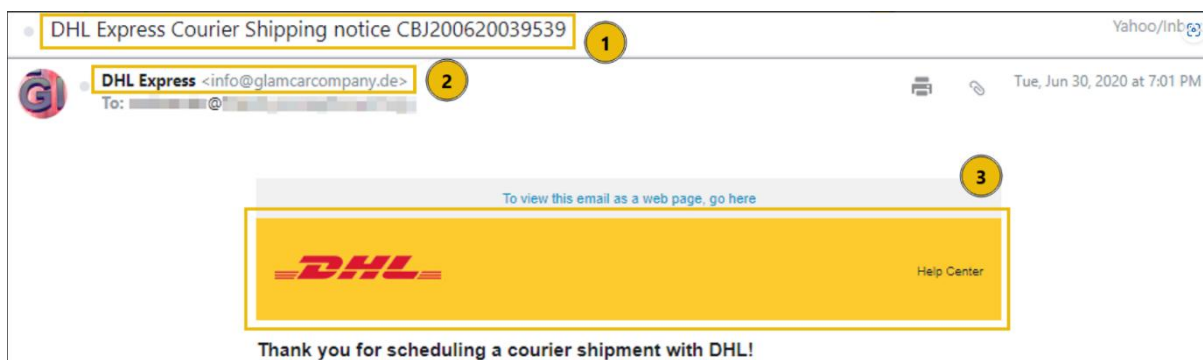
- The display name is set as "DHL Express"
- The actual sender email does not belong to the legitimate DHL domain
- The technique is used to gain the victim's trust

Brand Impersonation:

- DHL logos and HTML formatting are used
- The email visually resembles a legitimate DHL notification

Malicious Attachment:

- The email contains an Excel attachment
- Opening the attachment may execute malicious code or download additional malware.



4. Indicators of Compromise (IOCs)

IOC Type	Value
Display Name	DHL Express
Brand Used	DHL
Attachment Type	Microsoft Excel Document
Attack Category	Phishing
Delivery Method	Email Attachment

The image shows two side-by-side screenshots. The left screenshot is an email from DHL Express. It has a yellow header with the DHL logo and a 'Help Center' link. The body text says: 'Thank you for scheduling a courier shipment with DHL!', 'See attached for confirmation BL', 'CBJ200620039539', 'Reception', and an 'Important' section with instructions on packaging and pickup. At the bottom, there is a small icon of an Excel file labeled 'XLSX Attachment'.

The right screenshot is an Excel spreadsheet. The top row has columns A, B, and C. Column A contains 'To:' and 'MENON AND MENON LIMITED'. Column B contains 'Wanmangar, KOLHAPUR'. Column C contains a yellow box with the text 'TO TRANSLATE LANGUAGE KINDLY CLICK ENABLE EDITING ABOVE' and a 'Hyperlink' button. Below this, there is a table with 6 rows and 2 columns. The first column is labeled '表判:' and the second column is labeled 'Ref. No. PURNA/F34'. The rows contain the following text: 1. 姓名和地址, 2. 公司事务所, 3. 电话号码, 4. 传真号码, 5. 电子邮箱, 6. 设立日期, 7. 所有者及董事 M.D., 8. 重要的对应主要人, 9. 股东人, 10. 我市场的场域.

5. Analysis Findings

Subject Analysis:

The subject line creates urgency and encourages the recipient to review shipment details

Sender Analysis:

The display name appears legitimate; however, the underlying email address does not match DHL's official domain

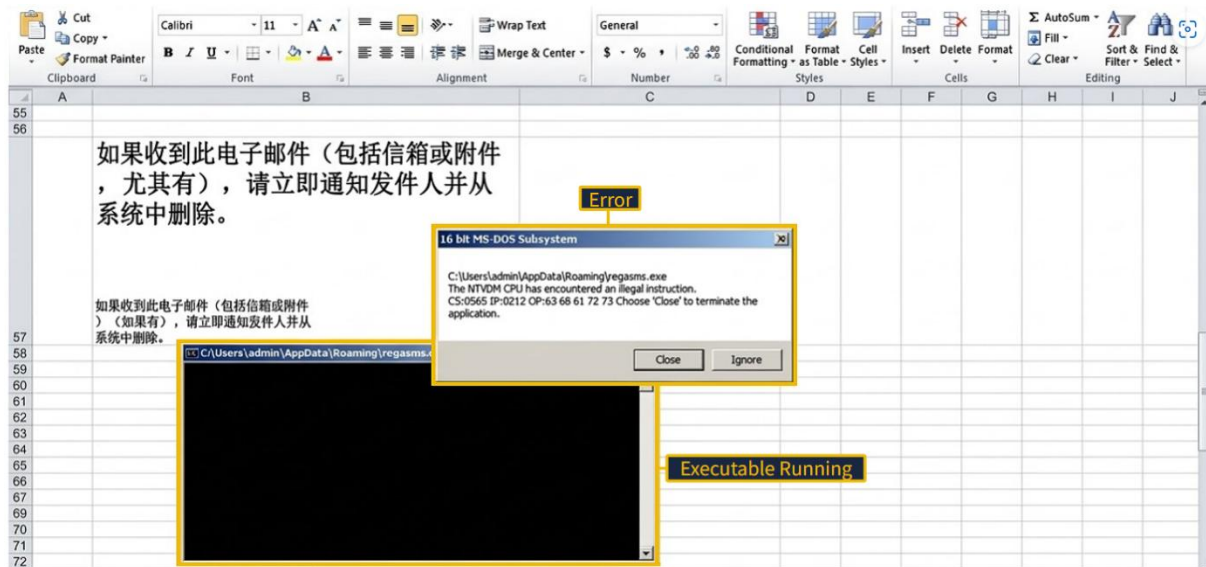
Content Analysis:

The email uses professional branding and formatting to appear authentic.

Attachment Analysis:

The attached Excel document may contain or trigger malicious activity. Such files commonly contain:

- Malicious macros
- Embedded scripts
- Malware downloaders
- Remote access trojans (RATs)



6. Detection Opportunities

- Detect suspicious attachment types.
- Scan attachments in a sandbox.
- Alert on brand impersonation.

7. Risk Assessment

Risk Level: High

Severity: High

Confidence: High

Category: Phishing / Malware Delivery

Reasons:

- Trusted brand impersonation
- Malicious attachment delivery
- Potential malware execution
- Possibility of credential theft or system compromise

8. MITRE ATT&CK Mapping

Technique ID : T1566.001

Technique Name : Phishing : Spearphishing Attachment

The attacker delivers a malicious Excel attachment through email to persuade the victim to open the file, potentially leading to malware execution or credential compromise.

9. Recommended Actions

1. Do not open the attachment.
2. Verify shipment notification through official DHL channels.
3. Block the sender domain
4. Submit the attachment to a sandbox environment
5. Update email security controls
6. Educate users about phishing indicators

10. Conclusion

This email is a phishing attempt that abuses DHL branding to deliver a malicious Excel attachment. Multiple phishing indicators, including sender impersonation, brand impersonation, and attachment-based malware delivery, support classifying the email as malicious and blocking it.